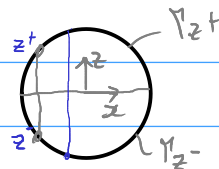
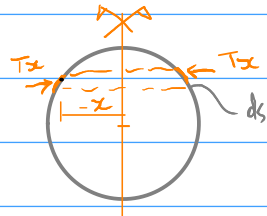


$\vec{T} = \begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix}$
 $\vec{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$
 $\vec{T} = \vec{\nabla} \cdot \vec{\eta} \rightarrow \text{normal a la esfera}$

$T_x = -\frac{x}{a} p_0 + \frac{3}{2} \mu \frac{U}{a}$; $T_y = -\frac{y}{a} p_0$; $T_z = -\frac{z}{a} p_0$

$\vec{F} = \int_r \vec{T} ds = \begin{bmatrix} \int_r T_x ds \\ \int_r T_y ds \\ \int_r T_z ds \end{bmatrix}$



$$F_z = \int_r T_z ds = -\frac{p_0}{a} \int_r z ds = -\frac{p_0}{a} \left(\int_{z+} z ds + \int_{z-} z ds \right) = 0$$

$$F_y = \int_r T_y ds = -\frac{p_0}{a} \int_r y ds = -\frac{p_0}{a} \cdot 0 = 0$$

$$\begin{aligned}
 F_x &= \int_r T_x ds = \int_r \left(-\frac{x}{a} p_0 + \frac{3}{2} \mu \frac{U}{a} \right) ds = \frac{3}{2} \mu \frac{U}{a} \int_r 1 ds \\
 &= \frac{3}{2} \mu \frac{U}{a} \overset{\text{superficie de la esfera}}{4\pi a^2} = 6\mu U \pi a = \underline{6\pi a \mu U}
 \end{aligned}$$